



FUZZY SETS, LOGIC AND SYSTEMS & APPLICATIONS

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INTENDED AUDIENCE : UG, PG Students, industry professionals, researchers etc.

COURSE OUTLINE :

The course is designed to give a solid grounding of fundamental concepts of fuzzy logic and its applications. The level of the course is chosen to be such that all students aspiring to be a part of computational intelligence directly or indirectly in near future should get these concepts.

ABOUT INSTRUCTOR :

Dr. Nishchal Kumar Verma is a Professor in the Department of Electrical Engineering at the Indian Institute of Technology Kanpur. He has received several awards, including the Smt. Lata and K.G. Karandikar Faculty Chair Professor (2024), the Teaching Excellence (2024), the prestigious research fellowship from UNEC, Azerbaijan (2023), a research fellowship from University of Tennessee, Memphis, USA (2022, 2010), and the Devendra Shukla Young Faculty Research Fellowship (2013). Dr. Verma specializes in the theoretical developments of Artificial Intelligence, Machine Learning, and Deep Learning, applying these concepts to Electrical Engineering and various interdisciplinary fields. His research areas include Large Language Models, Intelligent Instrumentation, Control, Automation, Computer Vision, Prognosis and Health Management of Machines, Bioinformatics, and Cyber-Physical Systems, among others. Dr. Verma and his team have developed several AI-based technologies for the BOEING Company, USA, and organized numerous national and international workshops, conferences, seminars, brainstorming sessions, and short-term courses. He has published over 270 research papers in reputable peer-reviewed journals and conferences and has authored, co-authored, or edited six books. He has successfully completed a significant number of research projects funded by various agencies, including The BOEING Company, DST, DRDO, MHRD, MeitY, CSIR, IIT Kanpur, MCIT, SFTIG, and VTOL etc. With over 20 years of experience in the fields of Artificial Intelligence and Machine Learning, Dr. Verma serves as an Associate Editor for key publications, including IEEE Transactions on Artificial Intelligence (2020-2024), IEEE Transactions on Neural Networks and Learning Systems (2019-2024), and IEEE Computational Intelligence Magazine (2015-2020).

COURSE PLAN :

Week 1: Introduction and Fuzzy Sets Theory

Week 2: Membership Functions

Week 3: Set Theoretic Operations

Week 4: Fuzzy Arithmetic

Week 5: Fuzzy Relations

Week 6: Fuzzy Inference Systems I

Week 7: Fuzzy Inference Systems II

Week 8: Wang and Mendel Model

Week 9: TSK Model

Week 10: Fuzzifiers and Defuzzifiers

Week 11: ANFIS Architecture

Week 12: Fuzzy Systems and Machine Learning